

Compiling by exact queries over index sets

Supplementary Information

Alberto Hernández-Espinosa

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S1 Cube algebra and queries

This section completes the algebra of Section 4 of the main text. Cubes, covers and their notation are as defined there.

Proposition S1 (Exact disjoint difference). *Subtracting a cube d from a cube c of the same width yields an exact disjoint cover of $c \setminus d$ with at most n cubes when they intersect, and c itself when they do not.*

Proof. Consider, in ascending order, the coordinates that are free in c but fixed in d . At each, emit the part of the current cube that takes the value opposite to d , and continue with the part that agrees. Emitted parts first disagree with d at different coordinates and are therefore disjoint. Every member of $c \setminus d$ has exactly one first such disagreement, and the final agreeing part lies inside d and is discarded. There are at most n split coordinates; with none, $c \subseteq d$ and the result is empty. $\square$

Restricting a coordinate is intersection with that coordinate's literal cube. An integer interval of a field is decomposed into aligned binary blocks, each a cube over the field with all other coordinates free. Covers may overlap, so their sizes cannot in general be added to obtain the size of a union.

Proposition S2 (Query soundness and conditional completeness). *If every leaf denotes its predicate exactly, every cube returned by the depth-first solver is contained in the denotation of the query. If the finite search is exhausted without a resource limit, the absence of an accepting branch proves the query unsatisfiable in its declared universe.*

Proof. The current cube starts as the universe. A leaf intersection keeps exactly the indices that satisfy the leaf; AND requires every child and OR enumerates the union of its children. By induction

over the pending expression each branch preserves these set identities, so a completed branch contains only satisfying indices, and exhausting all branches accounts for the whole expression. $\square$

The first depth-first witness is not guaranteed to be the globally smallest index; the first answer of the main text uses complete covers and their smallest anchors, which Proposition 1 of the main text makes exact. A timeout or a count limit is reported as unknown, never as unsatisfiable, and unsatisfiability refers only to the declared domains and the fixed context.

S2 A representation limit: parity

Proposition S3 (Parity needs an exponential flat cube cover). *For $n \geq 1$, every exact union of cubes equal to the set of odd-parity indices consists of 2^{n-1} singleton cubes.*

Proof. A cube with a free coordinate contains two indices that differ only there, and they have opposite parity, so it is not a subset of the odd-parity set. In an exact union every cube must be such a subset, since a union cannot cancel an unwanted member. Every cube is therefore a singleton, and there are 2^{n-1} odd-parity indices. $\square$

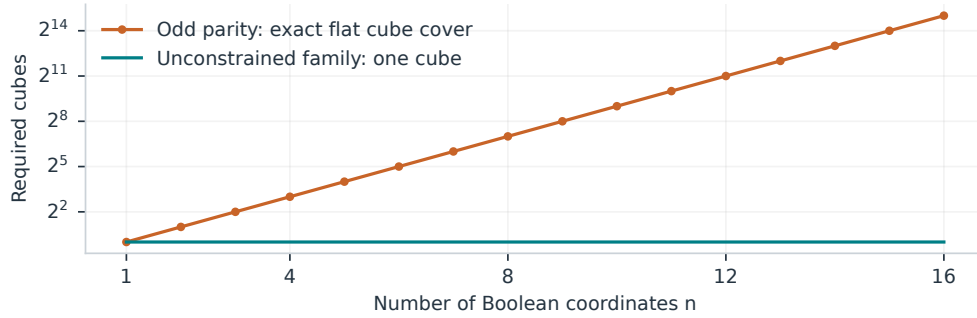


Figure S1: Analytic illustration, not a compiler measurement: an unconstrained family is one cube, while an exact flat cover of odd parity needs exponentially many. The bound is specific to flat unions of cubes; it does not establish the same growth for factored expressions or decision diagrams.

A cube with k free coordinates describes 2^k indices with two integers, but that is a saving of representation, not 2^k answers computed at constant cost; expanding the members takes at least their number, and for unbounded widths the integer operations themselves scale with the number of machine words. Compression must therefore be stated for a named representation and a named family of predicates, and the parity example is why.

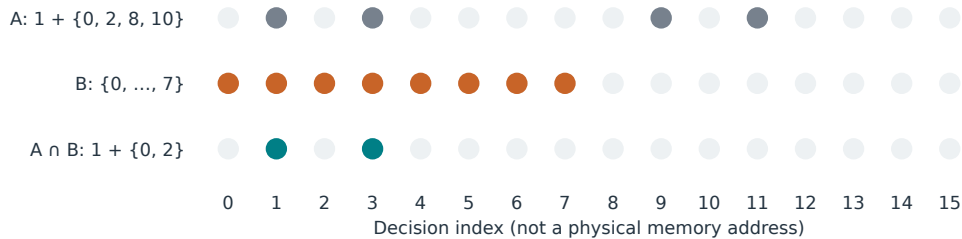


Figure S2: A four-coordinate example of the cube algebra: anchor 1 with free mask 10 denotes 1 plus each of $\{0, 2, 8, 10\}$; intersecting it with the indices up to 7 leaves 1 plus each of $\{0, 2\}$, which is again a cube. The dots expand the sets for reading.

S3 Round one: the ladder

Round one fixed five solvers before measuring anything, each adding one mechanism to the one below, so that adjacent arms form an ablation. A0 is the structural-encoding search of Phase 2; A1 removes its query cap; A2 replaces the target rectangles by one strict-product domain per window [3]; A3 adds four certified propagation rules [4]; A4 adds a multiscale catalogue of related windows [1] and a discrepancy-ordered traversal [2]. The arm with the largest mean paired $\log(J_{A0}/J_{\text{arm}})$ at 0.1 s on 100 development programs was selected (A4, development effect +0.0663), frozen, and then measured on 200 fresh programs.

Step (200 fresh, 0.1 s)	Mean log J ratio	95% interval	W / T / L	Time ratio
A1 over A0	+0.0055	[+0.0025, +0.0091]	18 / 182 / 0	1.17
A2 over A1	−0.0011	[−0.0028, +0.0000]	0 / 198 / 2	0.82
A3 over A2	+0.0000	[+0.0000, +0.0000]	0 / 200 / 0	0.83
A4 over A3	+0.0579	[+0.0442, +0.0726]	93 / 107 / 0	14.57

Table S1: Each step against the one below it. Positive log ratios mean the step lowered J . Time ratios are geometric means of per-program median compile times, step over previous.

A2 did not raise quality because the gaps in the target rectangles, though real in geometry, rarely held improvements on this generator; it did replace a heuristic refusal by a proof of emptiness and made each query cheaper. A3 changed no answer at all, which is what a sound propagator should do when every query already finishes: it can change when an answer is found but not which. The whole gain of the round is the last step, which bundles two mechanisms; round two took them apart. The primary comparisons of the round, with Bonferroni-adjusted 98.33% intervals, are in Table 6 of the main text; A4 had the best average output quality among A0, the accepted Phase 1 optimiser and the classical compiler at 0.1 s on that population, but lost to classical on 20 programs and took 76.20 times its compile time.

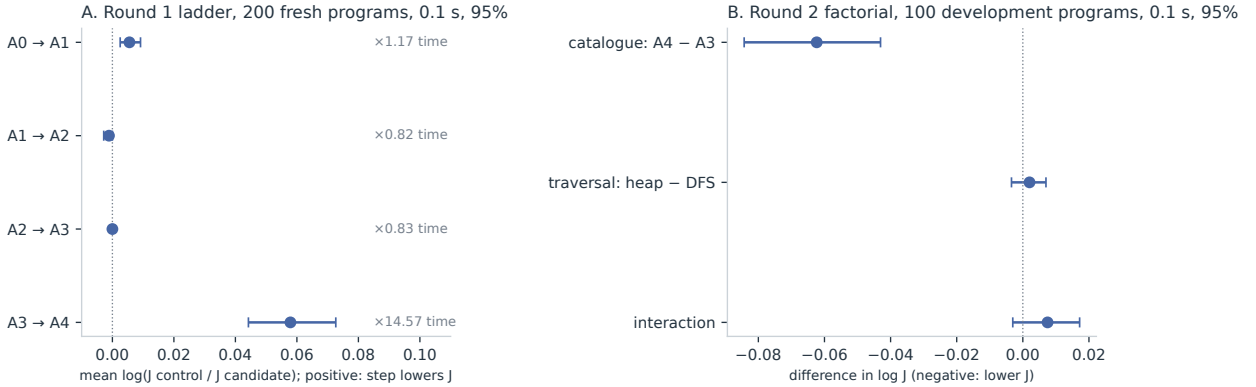


Figure S3: (A) The round-one ladder, each step against the one below it, with its compile-time ratio. (B) The round-two factorial on development programs: the catalogue lowers log J and its interval excludes zero; the traversal and the interaction intervals span zero. The generator checks both statements before rendering.

S4 Round two: repair, reversal and the factorial

Review of round one found that interrupted candidate validations were not counted when no learner was attached; decisions were unaffected, but the report said zero where the truth was one or more. The repair lives in a versioned successor that makes the same decisions node for node on a deterministic clock.

A reversal. On 200 fresh programs the repaired A4 beat the earlier `cap512_wider` optimiser (the Phase 2 controller with a larger query cap and wider windows): mean log J ratio $+0.0312$, 95% interval $[+0.0182, +0.0460]$, J ratio 0.969, with 55 programs better, 123 equal and 22 worse, at 0.978 times its compile time. On the public suite the order reversed: 2.102747 for `cap512_wider` against 2.088945 for A4. Both results are exact; one concerns a generator population and the other eight fixed programs.

The factorial. Crossing catalogue (A3’s small windows or A4’s multiscale catalogue) with traversal (depth-first or discrepancy heap) on 100 development programs gave a catalogue effect of -0.0623 in log J ($[-0.0843, -0.0430]$), a traversal effect of $+0.0020$ ($[-0.0034, +0.0070]$) and an interaction of $+0.0075$ ($[-0.0030, +0.0172]$). The catalogue explains the gain; the traversal is not shown to contribute, although its interval does not exclude a small effect.

The selected candidate. The development rule selected `cell_a4cat_dfs`, A4’s catalogue with depth-first search. On the fresh cohort its J ratio against heap A4 was 0.989 (98.75% interval $[0.978, 0.9997]$) and against `cap512_wider` 0.959. Both registered routes to a “better” claim required either an upper J bound of at most 0.98 or an upper compile-time bound of at most 0.80 at no loss of quality; neither was met, and the verdict was `TARGET_NOT_REACHED`. The candidate became R0 after the assurance round.

Learning. A depth-three schema tree over the code’s own bits [6], used as a guide for exact search in the spirit of [5], was compared with Hamming order, random order and a tree fitted on shuffled labels. It beat random order (0.061) and shuffled labels (0.045, below the registered minimum), but lost to Hamming order on all 30 of 30 fixtures (-0.107). Inside real compilations, not one of 100 programs had a query that reached the required number of validated observations, and only 33 of 1,639 learner queries reached the decision point: `ECONOMICALLY_UNAVAILABLE`. The review showed why: under sound bounds only strict improvements reach the observation hook, and the first accepted improvement ends the epoch, so at most one observation can exist when the learner is first prepared. Round one’s learning hypothesis had failed by design, since none of its 30 fixtures contained a test solution better than the training minimum. The per-query learner is retired for this architecture; other learning architectures are not ruled out.

Engineering. Propagation became about three-quarters of profiled compile time with the multiscale catalogue. A worklist fixed point saved 4.8% of compile time against a gate of 10.0%, and the parent was kept. The following assurance round proposed a filter on the capacity rule whose predicted fixed-work ratio was 0.939 (with a ceiling of 0.819 had the scan vanished entirely), against a gate of 0.80: `NO_JUSTIFIED_OPTIMIZATION`.

S5 The third round in detail

The mechanism decision. Before any integration, a cost model predicted C1’s fixed-work ratio from replays of captured propagation work. The registered formula gave a conservative 0.8268 against a gate of 0.80 and stopped the round. Review found that the subtracted term covered a narrower scope than the replayed work, so that overlapping work was charged twice, and that the adapter cost had been estimated with a loop that did not perform the real operations. The correction, admitted by the lead reviewer as post-hoc development accounting, re-measured the adapter in 90 fresh processes and gave a conservative 0.7864 (point 0.6061); the same scope correction with the old adapter gave 0.7638, and a replay-only sensitivity 0.7650. By family the corrected conservative forecasts were 0.831 (scalar), 0.721 (vector), 0.727 (mixed), 0.900 (dependency) and 0.767 (aliasing). These are forecasts on 30 development programs, not measurements or confidence bounds. The replay itself

reproduced R0’s outputs and certificate streams identically, with a median kernel speed-up of 2.3649 (range 0.920 to 4.361); an earlier summary had reported the upper median of the 30 values instead.

Development. On 100 development programs, 300 fixed-work pairs were identical, the fixed-work cost ratio was 0.589 against a gate of 0.90, and the mean log J ratio at 0.1 s was -0.0173 with 44 programs better, 256 equal and 0 worse. At equal wall time C1 completed more search (median 912 nodes against 471.5).

Confirmation by family. The cost ratio was 0.590 (scalar), 0.521 (vector), 0.536 (mixed), 0.642 (dependency) and 0.591 (aliasing); the quality ratio was 0.994, 0.987, 0.986, 0.995 and 0.991 in the same order. Median fixed-work compile calls were 0.187 s for C1 and 0.362 s for R0, with ninety-fifth percentiles of 1.07 s and 2.41 s. By search-size third, R0’s median compile call was 18.0 ms on the smallest third, 0.38 s on the middle third and 1.13 s on the largest; the summed-seconds ratio was 0.464. Other allowances: J ratio 0.986 at 0.01 s (150 better, 850 equal, 0 worse) and 0.997 at 1 s (36, 947, 17).

S6 Phase 1: the original direct compiler

The first version was accepted with limitations after an independent lead review of its implementation and evidence. Its verifier passed 351 recorded tests and its standalone export passed the acceptance corpus. On the public suite its score was 2.008466 against 1.901379 for classical, a composite-score advantage of 5.63% that already arises in the first answer; the bounded optimiser added nothing on the public programs and improved 6 of 100 generated programs. The score gain is not an execution speed-up and cannot be attributed to the representation alone, because the two compilers also use different heuristics.

Implementation refinements then reduced its full compile time by a factor of 2.132 relative to its previous version (paired 95% interval [2.127, 2.138], 15 repetitions per program), and bootstrap time by 1.163; an earlier run of the same code gave 1.947 and 1.134, so within-run intervals understate between-run variation. Compilation remained much slower than classical: the geometric means of per-program median candidate/classical ratios were 2.07 for bootstrap and 657.5 for full compilation.

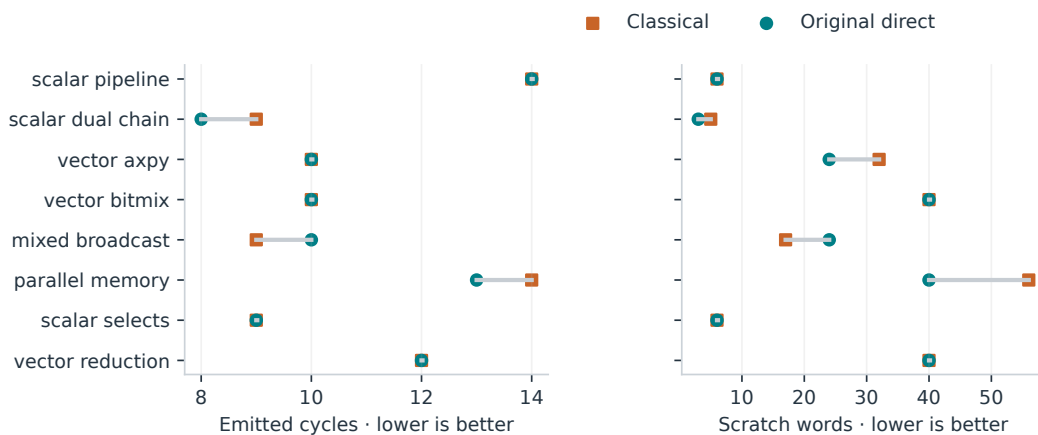


Figure S4: Phase 1, public programs: cycles and scratch words of the original direct compiler against classical. Lower is better on both axes.

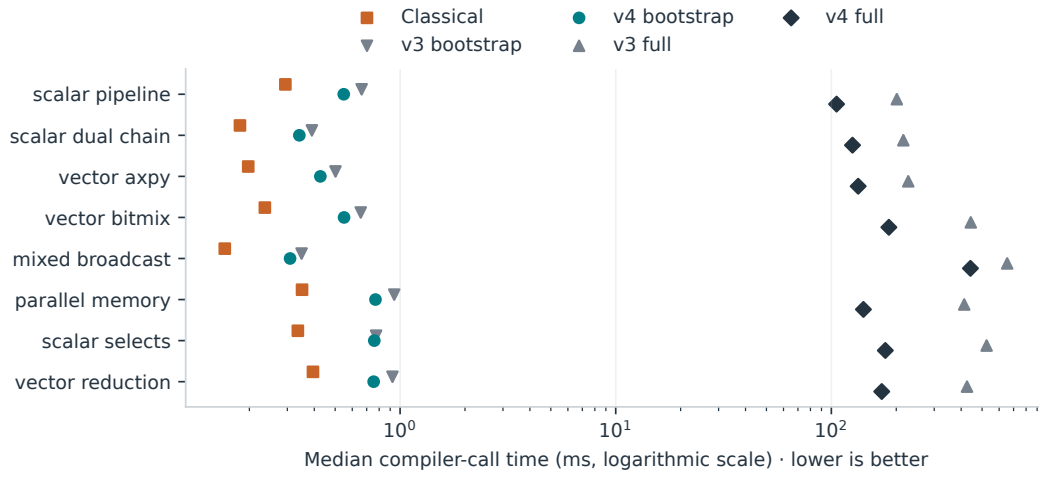


Figure S5: Phase 1: median compile-call times over 15 repetitions per public program for classical and both versions of the direct compiler.

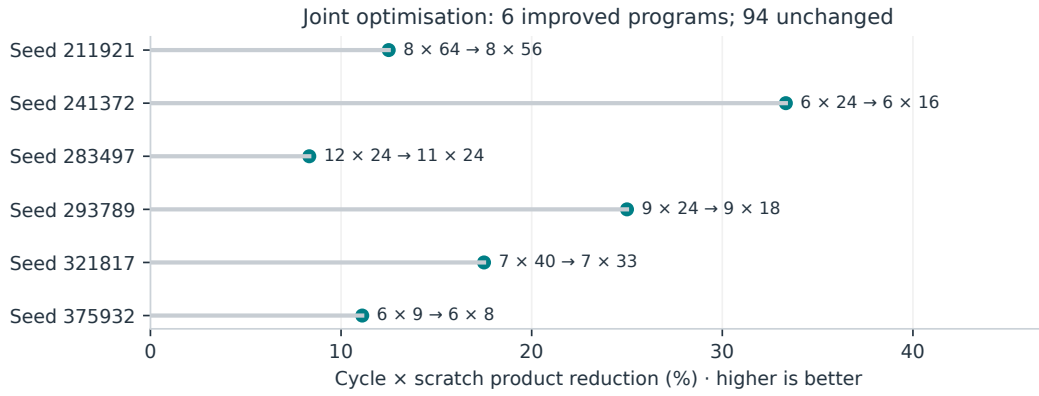


Figure S6: Phase 1: the generated programs improved by the bounded optimiser, with the product before and after.

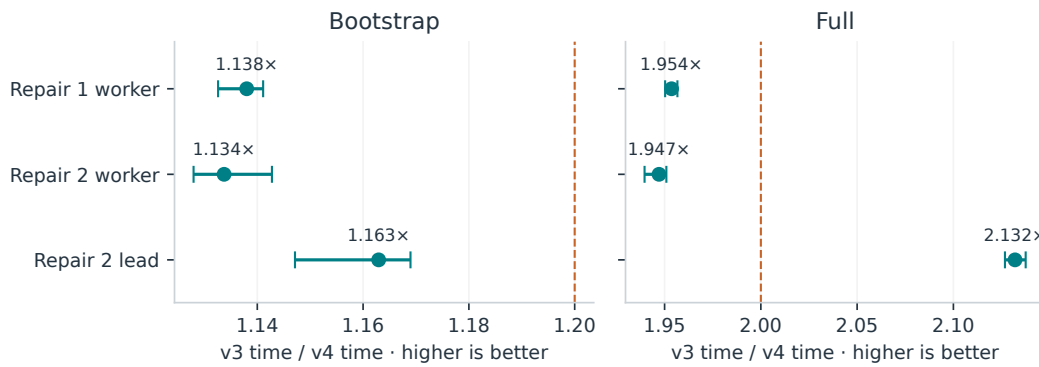


Figure S7: Phase 1: three runs of the same production code. Error bars are within-run paired intervals and do not describe variation between runs.

S7 Structural-encoding feasibility

A separate, pre-declared campaign asked whether exact structural encodings of whole-program choices could be sampled well enough to support later experiments. Finite checks covered 12 fixtures, 24 exhausted code universes and 564 round trips, with no failures. Two sampling streams per public program retained 160,000 attempts in 16 streams, 819 completed draws and 1,356 case checks with 0 discrepancies; raw-bit sampling produced no completions. The gate required 100 distinct completions per program, and 4 programs fell below it (Table S2). The campaign is inconclusive: this sampler did not pass its gate, which neither shows that structural encoding is impossible nor establishes any compression or performance benefit. The later stages were blocked and never measured.

Program	Raw-bit	Option-path	Distinct union	Gate
scalar pipeline	0	0	0	below minimum
scalar dual chain	0	234	234	meets minimum
vector axpy	0	194	194	meets minimum
vector bitmix	0	55	55	below minimum
mixed broadcast	0	135	135	meets minimum
parallel memory	0	16	16	below minimum
scalar selects	0	168	168	meets minimum
vector reduction	0	17	17	below minimum

Table S2: Completed draws per stream and the distinct union across streams, by public program.

S8 Review, errata and deviations

Self-review. The acceptance of C1 was performed by the implementer because the designated lead reviewer was unavailable; it is disclosed as a self-review, and an independent re-review would supersede it.

Erratum E1. The release handoff of the final round quotes a hash for C1 that matches no file or record in the repository; it was a transcription error. The measured C1 is the file with SHA-256 `b9697640e202c7cf2563ef3f11423f6cbd52403f91c897b62a6ab1b044672cfa` which appears in both freezes, the final source manifest and the export manifest.

Erratum E2. The handoff states that the frozen development auditor is an exact prefix of the current file. A later edit, made before the confirmation freeze, changed two places inside that prefix. Running the development audit from both files gives the same 32,362 checks in each case, with the frozen file’s own hash as the only finding, so the edit does not change the development audit.

Disclosed deviations. The kernel draft preceded its specification; an auditor was extended after the development freeze (E2); a false positive in the cohort inventory was corrected before any outcome; one unused variable was left in a frozen file to preserve its hash; development rows ran in a different order from confirmation rows; two auditor tests were re-targeted, not weakened; a few smoke runs were not charged to the ledger and are not evidence; and the cost-model correction of Section S5 is post hoc. None changes a gate.

Checks. The confirmation audit passed 143,585 checks with no findings and the repaired model audit 18,219. The review ran 82 tests of the final round and 53 inherited semantic tests, all passing, and checked 9,528 raw output streams against their rows.

S9 Artefacts and reproduction

The measured C1 source is `research/third_round_candidate.py`, with SHA-256

`b9697640e202c7cf2563ef3f11423f6cbd52403f91c897b62a6ab1b044672cfa`

R0 is `research/efficiency_search.py`. The evidence of the final round is under `results/phase2_structural_encoding/third_round_20260925_resume/` and its review under `results/phase2_structural_encoding/lead_resume_review_20260926/`. The paper is rebuilt, without any new timing, by

```
venv/bin/python luminal-challenge/paper/build_paper.py
```

which regenerates the figures, the tables and the value ledger, runs the worked example through the real compilers, re-derives every ledger entry from its source file, compiles both documents, and rejects unresolved references, overfull boxes and any digit in the manuscript that does not come from the ledger. The challenge asks that the exercise and its solution not be shared publicly, so these artefacts are kept local.

S10 The claim ledger

Every number in the main text and in this document is a macro whose value is recorded below with the file it comes from and the field or derivation that produces it. Pointer entries are re-read from disk and derived entries are recomputed on every build; entries computed by the generators (the worked example and the Phase 1 checks) are reproduced by rerunning them. Paths abbreviate `results/phase2_structural_encoding/` as `p2/`.

Key	Value	Source	Field or derivation
<code>aZeroAccHi</code>	0.0660	<code>p2/recovery_campaign_20260923_r3/COMPARISON.json</code>	<code>entries/16/quality_log_control_over_candidate/intervals/0.02</code>
<code>aZeroAccLo</code>	0.0251	<code>p2/recovery_campaign_20260923_r3/COMPARISON.json</code>	<code>entries/16/quality_log_control_over_candidate/intervals/0.02</code>
<code>aZeroAccLog</code>	0.0439	<code>p2/recovery_campaign_20260923_r3/COMPARISON.json</code>	<code>entries/16/quality_log_control_over_candidate/point_estimate</code>
<code>aZeroAccN</code>	100	<code>p2/recovery_campaign_20260923_r3/COMPARISON.json</code>	<code>entries/16/n_programs</code>
<code>aZeroAccRed</code>	4.3	<code>p2/recovery_campaign_20260923_r3/COMPARISON.json</code>	<code>entries/16/quality_log_control_over_candidate/point_estimate</code>
<code>aZeroAccTime</code>	0.073	<code>p2/recovery_campaign_20260923_r3/COMPARISON.json</code>	<code>entries/16/compile_candidate_over_control_ratio</code>
<code>aZeroAcclosses</code>	0	<code>p2/recovery_campaign_20260923_r3/COMPARISON.json</code>	<code>entries/16/losses</code>
<code>aZeroAccties</code>	74	<code>p2/recovery_campaign_20260923_r3/COMPARISON.json</code>	<code>entries/16/ties</code>
<code>aZeroAccwins</code>	26	<code>p2/recovery_campaign_20260923_r3/COMPARISON.json</code>	<code>entries/16/wins</code>
<code>aZeroClHi</code>	0.1194	<code>p2/recovery_campaign_20260923_r3/COMPARISON.json</code>	<code>entries/19/quality_log_control_over_candidate/intervals/0.02</code>
<code>aZeroClLo</code>	0.0592	<code>p2/recovery_campaign_20260923_r3/COMPARISON.json</code>	<code>entries/19/quality_log_control_over_candidate/intervals/0.02</code>
<code>aZeroClLog</code>	0.0882	<code>p2/recovery_campaign_20260923_r3/COMPARISON.json</code>	<code>entries/19/quality_log_control_over_candidate/point_estimate</code>
<code>aZeroClN</code>	100	<code>p2/recovery_campaign_20260923_r3/COMPARISON.json</code>	<code>entries/19/n_programs</code>
<code>aZeroClRed</code>	8.4	<code>p2/recovery_campaign_20260923_r3/COMPARISON.json</code>	<code>entries/19/quality_log_control_over_candidate/point_estimate</code>

Key	Value	Source	Field or derivation
aZeroClTime	7.286	p2/recovery_campaign_20260923_r3/C OMPARISON.json	entries/19/compile_candi date_over_control_ratio
aZeroCllosses	12	p2/recovery_campaign_20260923_r3/C OMPARISON.json	entries/19/losses
aZeroClties	35	p2/recovery_campaign_20260923_r3/C OMPARISON.json	entries/19/ties
aZeroClwins	53	p2/recovery_campaign_20260923_r3/C OMPARISON.json	entries/19/wins
aZeroPub	2.032760	p2/lead_release_review_20260924/PU BLIC_SCORE_RECOUNT.json	arms/structural_bound/sc ores/0
bA	0.01	plan/phase2_third_round/PROTOCOL.j son	wall_budgets_seconds/0
bB	0.1	plan/phase2_third_round/PROTOCOL.j son	wall_budgets_seconds/1
bC	1	plan/phase2_third_round/PROTOCOL.j son	wall_budgets_seconds/2
effCeil	0.819	p2/efficiency_20260925/MECHANISM_P ROPOSAL.json	prediction/zero_cost_cei ling_if_the_scan_vanished
effPred	0.939	p2/efficiency_20260925/MECHANISM_P ROPOSAL.json	prediction/fixed_work_co mpile_ratio_equal_family _predicted
exAnchor	1	generate_figures.pyschemaexample	anchor
exFree	10	generate_figures.pyschemaexample	freemask
exMembers	0,2,8,10	generate_figures.pyschemaexample	membersminusanchor
exRangeHi	7	generate_figures.pyschemaexample	intervalupperend
exResult	0,2	generate_figures.pyschemaexample	intersectionminusanchor
facCat	-0.0623	p2/next_round_20260925/FACTORIAL.j son	budgets/0.1/factorial/ca talog_effect_a4_minus_a3 /estimate
facCatHi	-0.0430	p2/next_round_20260925/FACTORIAL.j son	budgets/0.1/factorial/ca talog_effect_a4_minus_a3 /interval_95
facCatLo	-0.0843	p2/next_round_20260925/FACTORIAL.j son	budgets/0.1/factorial/ca talog_effect_a4_minus_a3 /interval_95
facInt	+0.0075	p2/next_round_20260925/FACTORIAL.j son	budgets/0.1/factorial/in teraction/estimate
facIntHi	+0.0172	p2/next_round_20260925/FACTORIAL.j son	budgets/0.1/factorial/in teraction/interval_95/1
facIntLo	-0.0030	p2/next_round_20260925/FACTORIAL.j son	budgets/0.1/factorial/in teraction/interval_95/0
facPrograms	100	p2/next_round_20260925/FACTORIAL.j son	budgets/0.1/factorial/pr ograms
facTrav	+0.0020	p2/next_round_20260925/FACTORIAL.j son	budgets/0.1/factorial/tr aversal_effect_heap_minu s_dfs/estima
facTravHi	+0.0070	p2/next_round_20260925/FACTORIAL.j son	budgets/0.1/factorial/tr aversal_effect_heap_minu s_dfs/interv
facTravLo	-0.0034	p2/next_round_20260925/FACTORIAL.j son	budgets/0.1/factorial/tr aversal_effect_heap_minu s_dfs/interv
feAttempts	160,000	p2/phase2_repair_20260923c/p1/summ ary.json	sampling_attempts_drawn
feBelow	4	p2/phase2_repair_20260923c/p1/summ ary.json	count(public_coverage[meets_minimum==false)
feCaseChecks	1,356	p2/phase2_repair_20260923c/p1/summ ary.json	sampling_case_checks
feCompletions	819	p2/phase2_repair_20260923c/p1/summ ary.json	sampling_completions
feDiscrepancies	0	p2/phase2_repair_20260923c/p1/summ ary.json	sampling_discrepancy_cou nt
feExhausted	24	p2/phase2_repair_20260923c/p1/summ ary.json	exhausted_comparisons
feFixtures	12	p2/phase2_repair_20260923c/p1/summ ary.json	len(fixtures)
feMinimum	100	p2/phase2_repair_20260923c/p1/summ ary.json	coverage_minimum
feRoundTrips	564	p2/phase2_repair_20260923c/p1/summ ary.json	p1_round_trips
feStreams	16	p2/phase2_repair_20260923c/p1/summ ary.json	stream_count

Key	Value	Source	Field or derivation
hC0ne	b9697640e202c7cf 2563ef3f11423f6c bd52403f91c897b6 2a6ab1b044672cfa	research/third_round_candidate.py	sha256offilebytes
hExpC0ne	58ee75059dcbdd9f	p2/third_round_20260925_resume/exp orts/C1/compiler.py	sha256offilebytes
hExpRZero	74c87b69e6306359	p2/third_round_20260925_resume/exp orts/R0/compiler.py	sha256offilebytes
hP0ne	d14bf39b450aaa5f	.build/direct_index/compiler.py	sha256offilebytes
hRZero	d0fcd441886cc5c3	research/efficiency_search.py	sha256offilebytes
ladFourHi	+0.0726	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/7/q uality/interval/1
ladFourLo	+0.0442	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/7/q uality/interval/0
ladFourQ	+0.0579	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/7/q uality/point
ladFourT	14.57	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/7/c ompile/geometric_ratio_c andidate_ove
ladFourlosses	0	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/7/q uality/losses
ladFourties	107	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/7/q uality/ties
ladFourwins	93	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/7/q uality/wins
ladOneHi	+0.0091	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/4/q uality/interval/1
ladOneLo	+0.0025	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/4/q uality/interval/0
ladOneQ	+0.0055	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/4/q uality/point
ladOneT	1.17	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/4/c ompile/geometric_ratio_c andidate_ove
ladOnelosses	0	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/4/q uality/losses
ladOneties	182	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/4/q uality/ties
ladOnewins	18	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/4/q uality/wins
ladThreeHi	+0.0000	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/6/q uality/interval/1
ladThreeLo	+0.0000	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/6/q uality/interval/0
ladThreeQ	+0.0000	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/6/q uality/point
ladThreeT	0.83	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/6/c ompile/geometric_ratio_c andidate_ove
ladThreelosses	0	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/6/q uality/losses
ladThreeties	200	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/6/q uality/ties
ladThreewins	0	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/6/q uality/wins
ladTwoHi	+0.0000	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/5/q uality/interval/1
ladTwoLo	-0.0028	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/5/q uality/interval/0
ladTwoQ	-0.0011	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/5/q uality/point
ladTwoT	0.82	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/5/c ompile/geometric_ratio_c andidate_ove
ladTwolosses	2	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/5/q uality/losses
ladTwoties	198	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/5/q uality/ties
ladTwowins	0	p2/objective_index_20260924/COMPAR ISON.json	ladder_unadjusted_95/5/q uality/wins
lrnBoundary	33	p2/next_round_20260925/LEARNING_FE ASIBILITY.json	economics/queries_reachi ng_boundary
lrnEcoPrograms	100	p2/next_round_20260925/LEARNING_FE ASIBILITY.json	economics/programs

Key	Value	Source	Field or derivation
lrnEcoZero	0	p2/next_round_20260925/LEARNING_FEASIBILITY.json	economics/programs_with_a_query_reaching_20_validated
lrnFixtures	30	p2/next_round_20260925/LEARNING_FEASIBILITY.json	evaluation_design/informative
lrnHam	-0.107	p2/next_round_20260925/LEARNING_FEASIBILITY.json	contrasts/tree_minus_hamming/estimate
lrnHamNeg	30	p2/next_round_20260925/LEARNING_FEASIBILITY.json	contrasts/tree_minus_hamming/negative
lrnQueries	1,639	p2/next_round_20260925/LEARNING_FEASIBILITY.json	economics/queries_with_learner
lrnRand	0.061	p2/next_round_20260925/LEARNING_FEASIBILITY.json	contrasts/tree_minus_random_mean/estimate
lrnShuf	0.045	p2/next_round_20260925/LEARNING_FEASIBILITY.json	contrasts/tree_minus_shuffled_tree/estimate
lvOne	98.33	plan/phase2_next_round/PROTOCOL.json	statistics/learning_three_contrasts_percentiles/0
lvStd	95	plan/phase2_next_round/PROTOCOL.json	statistics/primary_head_to_head_percentiles/0
lvThree	97.5	plan/phase2_third_round/PROTOCOL.json	statistics/percentiles/0
lvTwo	98.75	plan/phase2_next_round/PROTOCOL.json	statistics/new_candidate_four_endpoints_percentiles/0
mBits	32	.reference/machine.py	WORD_MASKbitwidth
mOneSlot	1	.reference/machine.py	ENGINE_LIMITSstore/flow
mPublic	8	.reference/programs	numberofpublicprograms
mTwoSlot	2	.reference/machine.py	ENGINE_LIMITSload/scalar/vector
mVlen	8	.reference/machine.py	VLEN
mWords	256	.reference/machine.py	SCRATCH_WORDS
pOneAccepted	18	results/direct_index_v4_optimization_repair2/lead_review/final/runs.json	extra_corpus/score_distribution/candidate_full/accepted Impr
pOneClassical	1.901379	results/direct_index_v4_optimization_repair2/lead_review/final/runs.json	analysis/scores/classical/combined_score
pOneExtraBoot	2.362098	results/direct_index_v4_optimization_repair2/lead_review/final/runs.json	extra_corpus/score_distribution/candidate_bootstrap/combined
pOneExtraCl	2.370890	results/direct_index_v4_optimization_repair2/lead_review/final/runs.json	extra_corpus/score_distribution/classical/combined_score_geo
pOneExtraFull	2.376597	results/direct_index_v4_optimization_repair2/lead_review/final/runs.json	extra_corpus/score_distribution/candidate_full/combined_score
pOneExtraN	100	results/direct_index_v4_optimization/baseline/extra_corpus_manifest.json	len(programs)
pOneGain	5.63	results/direct_index_v4_optimization_repair2/lead_review/final/runs.json	analysis.scorescandidate_full/classical-1
pOneImproved	6	results/direct_index_v4_optimization_repair2/lead_review/final/runs.json	extra_corpusruns: programswhosefullproduct bootstrap
pOneReps	15	results/direct_index_v4_optimization_repair2/lead_review/final/runs.json	runs:repetitionsper(arm, program), asserted
pOneScore	2.008466	results/direct_index_v4_optimization_repair2/lead_review/final/runs.json	analysis/scores/candidate_full/combined_score
pOneSlowBoot	2.07	results/direct_index_v4_optimization_repair2/lead_review/final/runs.json	classical_relative
pOneSlowFull	657.5	results/direct_index_v4_optimization_repair2/lead_review/final/runs.json	classical_relative
pOneSpeedBoot	1.163	results/direct_index_v4_optimization_repair2/lead_review/measurement_audit.json	speedups/bootstrap/point
pOneSpeedFull	2.132	results/direct_index_v4_optimization_repair2/lead_review/measurement_audit.json	speedups/full/point

Key	Value	Source	Field or derivation
pOneSpeedFullHi	2.138	results/direct_index_v4_optimization_repair2/lead_review/measurement_audit.json	speedups/full/high
pOneSpeedFullLo	2.127	results/direct_index_v4_optimization_repair2/lead_review/measurement_audit.json	speedups/full/low
pOneTests	351	results/direct_index_v4_optimization_repair2/lead_review/verification/summary.json	tests_total
pOneWorkerBoot	1.134	results/direct_index_v4_optimization_repair2/final/runs.json	analysis/ratios/bootstrap_candidate_vs_frozen/geometric_mean
pOneWorkerFull	1.947	results/direct_index_v4_optimization_repair2/final/runs.json	analysis/ratios/full_candidate_vs_frozen/geometric_mean_of_p
rOneAccHi	10.72	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/accepted_budgeted/interval/1
rOneAccLo	6.86	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/accepted_budgeted/interval/0
rOneAccRed	8.76	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/accepted_budgeted/geometric_J_ratio_control
rOneAccTime	0.82	p2/objective_index_20260924/COMPARISON.json	runtime_primary_bonferroni/accepted_budgeted: compile_seconds
rOneAccTimeHi	0.97	p2/objective_index_20260924/COMPARISON.json	runtime_primary_bonferroni/accepted_budgeted: compile_seconds
rOneAccTimeLo	0.68	p2/objective_index_20260924/COMPARISON.json	runtime_primary_bonferroni/accepted_budgeted: compile_seconds
rOneAccTimeThree	0.817	p2/objective_index_20260924/COMPARISON.json	runtime_primary_bonferroni/accepted_budgeted: compile_seconds
rOneAcclosses	0	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/accepted_budgeted/losses
rOneAccties	73	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/accepted_budgeted/ties
rOneAccwins	127	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/accepted_budgeted/wins
rOneAzeroHi	7.74	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/A0_frozen_phase2/interval/1
rOneAzeroLo	4.50	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/A0_frozen_phase2/interval/0
rOneAzeroRed	6.04	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/A0_frozen_phase2/geometric_J_ratio_control
rOneAzeroTime	11.65	p2/objective_index_20260924/COMPARISON.json	runtime_primary_bonferroni/A0_frozen_phase2: compile_seconds/
rOneAzeroTimeHi	13.53	p2/objective_index_20260924/COMPARISON.json	runtime_primary_bonferroni/A0_frozen_phase2: compile_seconds/
rOneAzeroTimeLo	10.03	p2/objective_index_20260924/COMPARISON.json	runtime_primary_bonferroni/A0_frozen_phase2: compile_seconds/
rOneAzerolosses	0	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/A0_frozen_phase2/losses
rOneAzeroties	98	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/A0_frozen_phase2/ties
rOneAzerowins	102	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/A0_frozen_phase2/wins
rOneBonf	99.17	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/classical/percentiles/1
rOneClHi	12.84	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/classical/interval/1
rOneClLo	8.38	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/classical/interval/0
rOneClRed	10.58	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/classical/geometric_J_ratio_control_over_c
rOneClTime	76.20	p2/objective_index_20260924/COMPARISON.json	runtime_primary_bonferroni/classical: compile_seconds/geometr

Key	Value	Source	Field or derivation
rOneClTimeHi	92.74	p2/objective_index_20260924/COMPARISON.json	runtime_primary_bonferro ni/classical:
rOneClTimeLo	62.41	p2/objective_index_20260924/COMPARISON.json	compile_seconds/interva runtime_primary_bonferro ni/classical:
rOneCllosses	20	p2/objective_index_20260924/COMPARISON.json	compile_seconds/interva primary/contrasts/classi cal/losses
rOneClties	42	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/classi cal/ties
rOneClwins	138	p2/objective_index_20260924/COMPARISON.json	primary/contrasts/classi cal/wins
rOneDevEffect	+0.0663	p2/objective_index_20260924/DEVELOPMENT.json	selection/best_effect
rOneLearnOf	30	p2/objective_index_20260924/LEARNING.json	design_diagnostic/of
rOneLearnZero	0	p2/objective_index_20260924/LEARNING.json	design_diagnostic/evalua tion_fixtures_with_a_tes t_object_bel
rOnePubAfour	2.088945	p2/objective_index_20260924/PUBLIC_SCORE.json	score/scores/arms/A4_mul tiscale_search@0.1/geome tric_mean
rOnePubAzero	2.032760	p2/objective_index_20260924/PUBLIC_SCORE.json	score/scores/arms/A0_fro zen_phase2@0.1/geometric _mean
rTwoDfsClL	15	p2/next_round_20260925/COMPARISON.json	descriptive_unadjusted_9 5/log(J_classical/J_cell _a4cat_dfs@0
rTwoDfsClLog	0.156	p2/next_round_20260925/COMPARISON.json	descriptive_unadjusted_9 5/log(J_classical/J_cell _a4cat_dfs@0
rTwoDfsClRed	14.4	p2/next_round_20260925/COMPARISON.json	descriptive_unadjusted_9 5/log(J_classical/J_cell _a4cat_dfs@0
rTwoDfsClT	39	p2/next_round_20260925/COMPARISON.json	descriptive_unadjusted_9 5/log(J_classical/J_cell _a4cat_dfs@0
rTwoDfsClW	146	p2/next_round_20260925/COMPARISON.json	descriptive_unadjusted_9 5/log(J_classical/J_cell _a4cat_dfs@0
rTwoDfsEHi	0.9757	p2/next_round_20260925/COMPARISON.json	candidate_endpoints/earl ier_cap512_wider/J_ratio _interval_98
rTwoDfsEJ	0.959	p2/next_round_20260925/COMPARISON.json	candidate_endpoints/earl ier_cap512_wider/J_ratio
rTwoDfsELo	0.939	p2/next_round_20260925/COMPARISON.json	candidate_endpoints/earl ier_cap512_wider/J_ratio _interval_98
rTwoDfsET	0.916	p2/next_round_20260925/COMPARISON.json	candidate_endpoints/earl ier_cap512_wider/compile _ratio
rTwoDfsEnL	14	p2/next_round_20260925/COMPARISON.json	candidate_endpoints/earl ier_cap512_wider/wins_tie s_losses/2
rTwoDfsEnT	116	p2/next_round_20260925/COMPARISON.json	candidate_endpoints/earl ier_cap512_wider/wins_tie s_losses/1
rTwoDfsEnW	70	p2/next_round_20260925/COMPARISON.json	candidate_endpoints/earl ier_cap512_wider/wins_tie s_losses/0
rTwoDfsHHi	0.9997	p2/next_round_20260925/COMPARISON.json	candidate_endpoints/cell _a4cat_heap/J_ratio_inte rval_98_75/1
rTwoDfsHJ	0.989	p2/next_round_20260925/COMPARISON.json	candidate_endpoints/cell _a4cat_heap/J_ratio
rTwoDfsHLo	0.978	p2/next_round_20260925/COMPARISON.json	candidate_endpoints/cell _a4cat_heap/J_ratio_inte rval_98_75/0
rTwoDfsHT	0.937	p2/next_round_20260925/COMPARISON.json	candidate_endpoints/cell _a4cat_heap/compile_ratio
rTwoDfsHnL	10	p2/next_round_20260925/COMPARISON.json	candidate_endpoints/cell _a4cat_heap/wins_ties_lo sses/2

Key	Value	Source	Field or derivation
rTwoDfsHnT	161	p2/next_round_20260925/COMPARISON.json	candidate_endpoints/cell_a4cat_heap/wins_ties_losses/1
rTwoDfsHnW	29	p2/next_round_20260925/COMPARISON.json	candidate_endpoints/cell_a4cat_heap/wins_ties_losses/0
rTwoERoute	0.80	plan/phase2_next_round/PROTOCOL.json	practical_targets/efficiency_route/upper_compile_ratio
rTwoHi	+0.0460	p2/next_round_20260925/COMPARISON.json	primary/interval_95/1
rTwoJ	0.969	p2/next_round_20260925/COMPARISON.json	primary/estimate
rTwoLo	+0.0182	p2/next_round_20260925/COMPARISON.json	primary/interval_95/0
rTwoLog	+0.0312	p2/next_round_20260925/COMPARISON.json	primary/estimate
rTwoPubCap	2.102747	p2/next_round_20260925/PUBLIC_SCORE.json	scores/arms/earlier_cap512_wider@0.1/min
rTwoPubClassical	1.901379	p2/next_round_20260925/PUBLIC_SCORE.json	scores/arms/classical@None/min
rTwoPubDfs	2.187957	p2/next_round_20260925/PUBLIC_SCORE.json	scores/arms/cell_a4cat_dfs@0.1/min
rTwoPubDirect	2.008466	p2/next_round_20260925/PUBLIC_SCORE.json	scores/arms/accepted_bootstrap@None/min
rTwoPubHeap	2.088945	p2/next_round_20260925/PUBLIC_SCORE.json	scores/arms/cell_a4cat_heap@0.1/min
rTwoQRoute	0.98	plan/phase2_next_round/PROTOCOL.json	practical_targets/quality_route/upper_J_ratio
rTwoTime	0.978	p2/next_round_20260925/COMPARISON.json	primary/compile_ratio_A4_over_earlier_geometric
rTwoWTLL	22	p2/next_round_20260925/COMPARISON.json	primary/wins_ties_losses_for_A4/2
rTwoWTLT	123	p2/next_round_20260925/COMPARISON.json	primary/wins_ties_losses_for_A4/1
rTwoWTLW	55	p2/next_round_20260925/COMPARISON.json	primary/wins_ties_losses_for_A4/0
rTwoWorklist	4.8	p2/next_round_20260925/ENGINEERING_DECISION.json	decision_by_budget/0.1/compile_time_reduction
rTwoWorklistGate	10.0	p2/next_round_20260925/ENGINEERING_DECISION.json	reason:gatethreshold
resident	8	plan/phase2_third_round/PROTOCOL.json	wall_limits/resident_queries
sliceMs	2	plan/phase2_third_round/PROTOCOL.json	wall_limits/slice_seconds
sliceNodes	2,048	plan/phase2_third_round/PROTOCOL.json	wall_limits/slice_nodes
tAccCases	277	p2/third_round_20260925_resume/stages/C_acceptance/rows.jsonl	acceptance
tAccPrograms	142	p2/third_round_20260925_resume/stages/C_acceptance/rows.jsonl	acceptance
tAdapterRows	90	p2/lead_resume_review_20260926/RECOMPUTE_M.json	adapter_rows
tAltSeedACostHi	0.598	p2/lead_resume_review_20260926/RECOMPUTE.json	endpoints/intervals_97_5_independent_draw_order/1/cost/1
tAltSeedBCostHi	0.597	p2/lead_resume_review_20260926/RECOMPUTE.json	endpoints/intervals_97_5_independent_draw_order/2/cost/1
tAltSeedCCostHi	0.598	p2/lead_resume_review_20260926/RECOMPUTE.json	endpoints/intervals_97_5_independent_draw_order/3/cost/1
tAuditC	143,585	p2/third_round_20260925_resume/checks/AUDIT_C.json	total_checks
tAuditD	32,362	p2/lead_resume_review_20260926/D_AUDIT_BOTH.json	frozen/total_checks
tAuditM	18,219	p2/third_round_20260925_resume/checks/AUDIT_M_final.json	total_checks
tCFamaliasing	0.591	p2/third_round_20260925_resume/COMPARISON.json	family/aliasing/cost_ratio
tCFamdependency	0.642	p2/third_round_20260925_resume/COMPARISON.json	family/dependency/cost_ratio
tCFammixed	0.536	p2/third_round_20260925_resume/COMPARISON.json	family/mixed/cost_ratio

Key	Value	Source	Field or derivation
tCFamscalar	0.590	p2/third_round_20260925_resume/COMPARISON.json	family/scalar/cost_ratio
tCFamvector	0.521	p2/third_round_20260925_resume/COMPARISON.json	family/vector/cost_ratio
tCertMedian	31,572	p2/third_round_20260925_resume/stages/C_fixed_work/rows.jsonl	certificate_median
tCost	0.574	p2/third_round_20260925_resume/COMPARISON.json	cost_ratio
tCostFour	0.5743	p2/third_round_20260925_resume/COMPARISON.json	cost_ratio
tCostHi	0.598	p2/third_round_20260925_resume/COMPARISON.json	intervals_97_5/cost/1
tCostHiFour	0.5978	p2/third_round_20260925_resume/COMPARISON.json	intervals_97_5/cost/1
tCostLo	0.552	p2/third_round_20260925_resume/COMPARISON.json	intervals_97_5/cost/0
tCostLoFour	0.5518	p2/third_round_20260925_resume/COMPARISON.json	intervals_97_5/cost/0
tDevCost	0.589	p2/third_round_20260925_resume/DEV ELOPMENT.json	fixed_work_compile_ratio_equal_family
tDevGate	0.90	p2/third_round_20260925_resume/DEV ELOPMENT.json	gate/fixed_work_compile_ratio_max
tDevLogJ	-0.0173	p2/third_round_20260925_resume/DEV ELOPMENT.json	wall_primary_mean_log_J_ratio_equal_family
tDevNodesC	912	p2/third_round_20260925_resume/DEV ELOPMENT.json	wall_nodes_median/C1
tDevNodesR	471.5	p2/third_round_20260925_resume/DEV ELOPMENT.json	wall_nodes_median/R0
tDevParity	300	p2/third_round_20260925_resume/DEV ELOPMENT.json	fixed_work_parity/pairs
tDevPrograms	100	p2/third_round_20260925_resume/DEV ELOPMENT.json	len(programs)
tDevWTLl	0	p2/third_round_20260925_resume/DEV ELOPMENT.json	wall_J_wins_ties_losses_C1/2
tDevWTLT	256	p2/third_round_20260925_resume/DEV ELOPMENT.json	wall_J_wins_ties_losses_C1/1
tDevWTLW	44	p2/third_round_20260925_resume/DEV ELOPMENT.json	wall_J_wins_ties_losses_C1/0
tDriftA	0.567	p2/lead_resume_review_20260926/REC OMPUTE.json	probes/drift_by_time_quartile/0
tDriftB	0.538	p2/lead_resume_review_20260926/REC OMPUTE.json	probes/drift_by_time_quartile/1
tDriftC	0.598	p2/lead_resume_review_20260926/REC OMPUTE.json	probes/drift_by_time_quartile/2
tDriftD	0.597	p2/lead_resume_review_20260926/REC OMPUTE.json	probes/drift_by_time_quartile/3
tExportRows	1,248	p2/third_round_20260925_resume/COMPARISON.json	stages/C_export/observed
tExportValid	1,248	p2/lead_resume_review_20260926/REC OMPUTE.json	checks/export_revalidated/valid
tFamilies	5	p2/lead_resume_review_20260926/REC OMPUTE.json	len(checks.family_sizes)
tFixedMedC1	0.187	p2/third_round_20260925_resume/COMPARISON.json	descriptive/fixed_work_C1/median
tFixedMedR0	0.362	p2/third_round_20260925_resume/COMPARISON.json	descriptive/fixed_work_R0/median
tFixedPctC1	1.07	p2/third_round_20260925_resume/COMPARISON.json	descriptive/fixed_work_C1/p95
tFixedPctR0	2.41	p2/third_round_20260925_resume/COMPARISON.json	descriptive/fixed_work_R0/p95
tGateCost	0.80	p2/third_round_20260925_resume/COMPARISON.json	gate/fixed_work_compile_ratio_upper_max
tGateQual	1.01	p2/third_round_20260925_resume/COMPARISON.json	gate/wall_primary_J_ratio_upper_max
tImport	0.601	p2/lead_resume_review_20260926/REC OMPUTE.json	probes/cost_including_import
tInherited	53	seederivation	log_count
tJrA	0.986	p2/third_round_20260925_resume/COMPARISON.json	descriptive/wall:0.01_J_ratio_geo
tJrB	0.991	p2/third_round_20260925_resume/COMPARISON.json	descriptive/wall:0.1_J_ratio_geo
tJrC	0.997	p2/third_round_20260925_resume/COMPARISON.json	descriptive/wall:1.0_J_ratio_geo

Key	Value	Source	Field or derivation
tKernelMax	4.361	p2/third_round_20260925_resume/REC ALIBRATED_PREDICTION.json	kernel_speedup_range/1
tKernelMed	2.3649	p2/third_round_20260925_resume/REC ALIBRATED_PREDICTION.json	kernel_speedup_median
tKernelMin	0.920	p2/third_round_20260925_resume/REC ALIBRATED_PREDICTION.json	kernel_speedup_range/0
tLossNoUnknown	10	p2/third_round_20260925_resume/sta ges/C_wall/rows.jsonl	wall_losses
tLossPrograms	4	p2/third_round_20260925_resume/sta ges/C_wall/rows.jsonl	wall_losses
tLossRuns	17	p2/third_round_20260925_resume/sta ges/C_wall/rows.jsonl	wall_losses
tLossUnknownA	0	p2/lead_resume_review_20260926/REC OMPLETE.json	probes/wall:0.01_losses_ with_unknown_queries
tLossUnknownB	6	p2/lead_resume_review_20260926/REC OMPLETE.json	probes/wall:0.1_losses_w ith_unknown_queries
tLossUnknownC	7	p2/lead_resume_review_20260926/REC OMPLETE.json	probes/wall:1.0_losses_w ith_unknown_queries
tMCons	0.7864	p2/third_round_20260925_resume/REC ALIBRATED_PREDICTION.json	corrected_conservative
tMFamaliasing	0.767	p2/third_round_20260925_resume/REC ALIBRATED_PREDICTION.json	family_corrected_conserv ative/aliasing
tMFamdependency	0.900	p2/third_round_20260925_resume/REC ALIBRATED_PREDICTION.json	family_corrected_conserv ative/dependency
tMFammixed	0.727	p2/third_round_20260925_resume/REC ALIBRATED_PREDICTION.json	family_corrected_conserv ative/mixed
tMFamscalar	0.831	p2/third_round_20260925_resume/REC ALIBRATED_PREDICTION.json	family_corrected_conserv ative/scalar
tMFamvector	0.721	p2/third_round_20260925_resume/REC ALIBRATED_PREDICTION.json	family_corrected_conserv ative/vector
tMGate	0.80	p2/third_round_20260925/MECHANISM_ DECISION.json	gate/conservative_ratio_ max
tMOldAdapter	0.7638	p2/third_round_20260925_resume/REC ALIBRATED_PREDICTION.json	corrected_scope_old_adap ter_conservative
tMPoint	0.6061	p2/third_round_20260925_resume/REC ALIBRATED_PREDICTION.json	corrected_point
tMReg	0.8268	p2/third_round_20260925/MECHANISM_ DECISION.json	gate/registered conserva tive_ratio
tMReplayOnly	0.7650	p2/third_round_20260925_resume/REC ALIBRATED_PREDICTION.json	replay_only_sensitivity_ conservative
tMWorkloads	30	p2/lead_resume_review_20260926/REC OMPLETE_M.json	programs
tMemMax	1.302	p2/third_round_20260925_resume/DEV ELOPMENT.json	memory_peak_ratio_max
tMemMed	1.051	p2/third_round_20260925_resume/DEV ELOPMENT.json	memory_peak_ratio_median
tMemPrograms	30	p2/third_round_20260925_resume/sta ges/D_memory/rows.jsonl	d_memory_programs
tNodes	10,000	plan/phase2_third_round/PROTOCOL.j son	fixed_work/aggregate_nod es
tOrderC1	0.577	p2/lead_resume_review_20260926/REC OMPLETE.json	probes/arm_order/C1/geo_ ratio
tOrderC1Pairs	461	p2/lead_resume_review_20260926/REC OMPLETE.json	probes/arm_order/C1/pairs
tOrderR0	0.572	p2/lead_resume_review_20260926/REC OMPLETE.json	probes/arm_order/R0/geo_ ratio
tOrderR0Pairs	539	p2/lead_resume_review_20260926/REC OMPLETE.json	probes/arm_order/R0/pairs
tPPmax	1.076	p2/lead_resume_review_20260926/REC OMPLETE.json	probes/per_program_cost_ ratio/max
tPPmedian	0.554	p2/lead_resume_review_20260926/REC OMPLETE.json	probes/per_program_cost_ ratio/median
tPPmin	0.264	p2/lead_resume_review_20260926/REC OMPLETE.json	probes/per_program_cost_ ratio/min
tPPq10	0.427	p2/lead_resume_review_20260926/REC OMPLETE.json	probes/per_program_cost_ ratio/q10
tPPq90	0.878	p2/lead_resume_review_20260926/REC OMPLETE.json	probes/per_program_cost_ ratio/q90
tPPslower	7	p2/lead_resume_review_20260926/REC OMPLETE.json	probes/per_program_cost_ ratio/programs_C1_slower
tParityMismatch	0	p2/third_round_20260925_resume/COM PARISON.json	fixed_work_parity/mismat ch_count
tParityPairs	1,000	p2/third_round_20260925_resume/COM PARISON.json	fixed_work_parity/pairs

Key	Value	Source	Field or derivation
tPctHi	98.75	p2/third_round_20260925_resume/COMPARISON.json	intervals_97_5/percentiles/1
tPctLo	1.25	p2/third_round_20260925_resume/COMPARISON.json	intervals_97_5/percentiles/0
tPerFamily	40	p2/lead_resume_review_20260926/RECOMPUTE.json	checks/family_sizes/vector
tProcess	0.649	p2/lead_resume_review_20260926/RECOMPUTE.json	probes/process_seconds_ratio
tProgBetter	25	p2/lead_resume_review_20260926/RECOMPUTE.json	probes/quality_programs_better_equal_worse/0
tProgEqual	173	p2/lead_resume_review_20260926/RECOMPUTE.json	probes/quality_programs_better_equal_worse/1
tProgWorse	2	p2/lead_resume_review_20260926/RECOMPUTE.json	probes/quality_programs_better_equal_worse/2
tPrograms	200	p2/third_round_20260925_resume/COMPARISON.json	programs
tPubC1A	2.135	p2/third_round_20260925_resume/COMPARISON.json	descriptive/public_scores/C1@wall:0.01/0
tPubC1B	2.188	p2/third_round_20260925_resume/COMPARISON.json	descriptive/public_scores/C1@wall:0.1/0
tPubC1C	2.188	p2/third_round_20260925_resume/COMPARISON.json	descriptive/public_scores/C1@wall:1.0/0
tPubROA	2.081	p2/third_round_20260925_resume/COMPARISON.json	descriptive/public_scores/R0@wall:0.01/0
tPubROB	2.188	p2/third_round_20260925_resume/COMPARISON.json	descriptive/public_scores/R0@wall:0.1/0
tPubROC	2.188	p2/third_round_20260925_resume/COMPARISON.json	descriptive/public_scores/R0@wall:1.0/0
tPubSixC1A	2.135275	p2/third_round_20260925_resume/COMPARISON.json	descriptive/public_scores/C1@wall:0.01/0
tPubSixC1B	2.187957	p2/third_round_20260925_resume/COMPARISON.json	descriptive/public_scores/C1@wall:0.1/0
tPubSixC1C	2.187957	p2/third_round_20260925_resume/COMPARISON.json	descriptive/public_scores/C1@wall:1.0/0
tPubSixROA	2.081045	p2/third_round_20260925_resume/COMPARISON.json	descriptive/public_scores/R0@wall:0.01/0
tPubSixROB	2.187957	p2/third_round_20260925_resume/COMPARISON.json	descriptive/public_scores/R0@wall:0.1/0
tPubSixROC	2.187957	p2/third_round_20260925_resume/COMPARISON.json	descriptive/public_scores/R0@wall:1.0/0
tPytest	82	seederivation	log_count
tQFamaliasing	0.991	p2/third_round_20260925_resume/COMPARISON.json	family/aliasing/quality_ratio
tQFamdependency	0.995	p2/third_round_20260925_resume/COMPARISON.json	family/dependency/quality_ratio
tQFammixed	0.986	p2/third_round_20260925_resume/COMPARISON.json	family/mixed/quality_ratio
tQFamscalar	0.994	p2/third_round_20260925_resume/COMPARISON.json	family/scalar/quality_ratio
tQFamvector	0.987	p2/third_round_20260925_resume/COMPARISON.json	family/vector/quality_ratio
tQoneSeed	980183	p2/third_round_20260925_resume/stages/C_wall/rows.jsonl	seedwiththelargestJgapamongthelosses(namedinR
tQual	0.991	p2/third_round_20260925_resume/COMPARISON.json	quality_ratio
tQualFour	0.9908	p2/third_round_20260925_resume/COMPARISON.json	quality_ratio
tQualHi	0.996	p2/third_round_20260925_resume/COMPARISON.json	intervals_97_5/quality/1
tQualHiFour	0.9964	p2/third_round_20260925_resume/COMPARISON.json	intervals_97_5/quality/1
tQualLo	0.985	p2/third_round_20260925_resume/COMPARISON.json	intervals_97_5/quality/0
tQualLoFour	0.9846	p2/third_round_20260925_resume/COMPARISON.json	intervals_97_5/quality/0
tRawStdout	9,528	p2/lead_resume_review_20260926/RECOMPUTE.json	checks/raw_stdout/checked
tReps	5	p2/third_round_20260925_resume/stages/C_fixed_work/rows.jsonl	reps_per_program
tResamples	10,000	p2/third_round_20260925_resume/COMPARISON.json	intervals_97_5/resamples
tSeed	2026092802	p2/third_round_20260925_resume/COMPARISON.json	intervals_97_5/seed

Key	Value	Source	Field or derivation
tSeed980026C1J	1,440	p2/third_round_20260925_resume/stages/C_wall/rows.jsonl	wall_seed
tSeed980026C1cycles	20	p2/third_round_20260925_resume/stages/C_wall/rows.jsonl	wall_seed
tSeed980026C1scratch	72	p2/third_round_20260925_resume/stages/C_wall/rows.jsonl	wall_seed
tSeed980026R0J	1,280	p2/third_round_20260925_resume/stages/C_wall/rows.jsonl	wall_seed
tSeed980026R0cycles	20	p2/third_round_20260925_resume/stages/C_wall/rows.jsonl	wall_seed
tSeed980026R0scratch	64	p2/third_round_20260925_resume/stages/C_wall/rows.jsonl	wall_seed
tSeed980183C1J	924	p2/third_round_20260925_resume/stages/C_wall/rows.jsonl	wall_seed
tSeed980183C1cycles	22	p2/third_round_20260925_resume/stages/C_wall/rows.jsonl	wall_seed
tSeed980183C1scratch	42	p2/third_round_20260925_resume/stages/C_wall/rows.jsonl	wall_seed
tSeed980183R0J	726	p2/third_round_20260925_resume/stages/C_wall/rows.jsonl	wall_seed
tSeed980183R0cycles	22	p2/third_round_20260925_resume/stages/C_wall/rows.jsonl	wall_seed
tSeed980183R0scratch	33	p2/third_round_20260925_resume/stages/C_wall/rows.jsonl	wall_seed
tSeedHi	980199	p2/lead_resume_review_20260926/REC OMPUTE.json	checks/cohort/seeds/1
tSeedLo	980000	p2/lead_resume_review_20260926/REC OMPUTE.json	checks/cohort/seeds/0
tSubstantive	975	p2/third_round_20260925_resume/stages/C_fixed_work/rows.jsonl	substantive_pairs
tSummed	0.464	p2/lead_resume_review_20260926/REC OMPUTE.json	probes/summed_median_seconds/ratio
tTerA	0.730	p2/lead_resume_review_20260926/REC OMPUTE.json	probes/cost_by_nodes_tercile/0/geo_ratio
tTerAhi	493	p2/lead_resume_review_20260926/REC OMPUTE.json	tercile
tTerAlo	0	p2/lead_resume_review_20260926/REC OMPUTE.json	tercile
tTerAms	18.0	p2/lead_resume_review_20260926/REC OMPUTE.json	probes/cost_by_nodes_tercile/0/R0_median_s
tTerB	0.527	p2/lead_resume_review_20260926/REC OMPUTE.json	probes/cost_by_nodes_tercile/1/geo_ratio
tTerBhi	9,999	p2/lead_resume_review_20260926/REC OMPUTE.json	tercile
tTerBlo	568	p2/lead_resume_review_20260926/REC OMPUTE.json	tercile
tTerBs	0.38	p2/lead_resume_review_20260926/REC OMPUTE.json	probes/cost_by_nodes_tercile/1/R0_median_s
tTerC	0.491	p2/lead_resume_review_20260926/REC OMPUTE.json	probes/cost_by_nodes_tercile/2/geo_ratio
tTerChi	9,999	p2/lead_resume_review_20260926/REC OMPUTE.json	tercile
tTerClo	9,999	p2/lead_resume_review_20260926/REC OMPUTE.json	tercile
tTerCs	1.13	p2/lead_resume_review_20260926/REC OMPUTE.json	probes/cost_by_nodes_tercile/2/R0_median_s
tTraces	196	p2/third_round_20260925_resume/stages/C_fixed_work/rows.jsonl	distinct_traces
tWTLAL	0	p2/third_round_20260925_resume/COMPARISON.json	descriptive/wall: 0.01_W_T_L_C1/2
tWTLAT	850	p2/third_round_20260925_resume/COMPARISON.json	descriptive/wall: 0.01_W_T_L_C1/1
tWTLAW	150	p2/third_round_20260925_resume/COMPARISON.json	descriptive/wall: 0.01_W_T_L_C1/0
tWTLBL	6	p2/third_round_20260925_resume/COMPARISON.json	descriptive/wall: 0.1_W_T_L_C1/2
tWTLBT	871	p2/third_round_20260925_resume/COMPARISON.json	descriptive/wall: 0.1_W_T_L_C1/1
tWTLBW	123	p2/third_round_20260925_resume/COMPARISON.json	descriptive/wall: 0.1_W_T_L_C1/0
tWTLCL	17	p2/third_round_20260925_resume/COMPARISON.json	descriptive/wall: 1.0_W_T_L_C1/2
tWTLCT	947	p2/third_round_20260925_resume/COMPARISON.json	descriptive/wall: 1.0_W_T_L_C1/1

Key	Value	Source	Field or derivation
tWTLCW	36	p2/third_round_20260925_resume/COMPARISON.json	descriptive/wall:1.0_W_T_L_C1/0
tWallTimeA	0.984	p2/lead_resume_review_20260926/RECOMPUTE.json	probes/wall:0.01_compile_call_time_geo_ratio
tWallTimeB	0.854	p2/lead_resume_review_20260926/RECOMPUTE.json	probes/wall:0.1_compile_call_time_geo_ratio
tWallTimeC	0.755	p2/lead_resume_review_20260926/RECOMPUTE.json	probes/wall:1.0_compile_call_time_geo_ratio
tZeroNode	5	p2/third_round_20260925_resume/stages/C_fixed_work/rows.jsonl	zero_node_programs
tabPPmixedbroadcastCOne	9x16	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	mixed_broadcast:COne
tabPPmixedbroadcastC1	9x17	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	mixed_broadcast:C1
tabPPmixedbroadcastDi	10x24	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	mixed_broadcast:Di
tabPPmixedbroadcastJ	144	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	mixed_broadcast:J
tabPPmixedbroadcastSer	11x42	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	mixed_broadcast:Ser
tabPPparallelmemoryCOne	13x40	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	parallel_memory:COne
tabPPparallelmemoryC1	14x56	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	parallel_memory:C1
tabPPparallelmemoryDi	13x40	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	parallel_memory:Di
tabPPparallelmemoryJ	520	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	parallel_memory:J
tabPPparallelmemorySer	21x128	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	parallel_memory:Ser
tabPPscalardualchainCOne	8x3	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	scalar_dual_chain:COne
tabPPscalardualchainC1	9x5	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	scalar_dual_chain:C1
tabPPscalardualchainDi	8x3	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	scalar_dual_chain:Di
tabPPscalardualchainJ	24	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	scalar_dual_chain:J
tabPPscalardualchainSer	12x9	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	scalar_dual_chain:Ser
tabPPscalarpipelineCOne	14x6	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	scalar_pipeline:COne
tabPPscalarpipelineC1	14x6	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	scalar_pipeline:C1
tabPPscalarpipelineDi	14x6	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	scalar_pipeline:Di
tabPPscalarpipelineJ	84	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	scalar_pipeline:J
tabPPscalarpipelineSer	18x15	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	scalar_pipeline:Ser
tabPPscalarselectsCOne	10x5	p2/next_round_20260925/PUBLIC_SCORE.jsonandp2/third_round_20260925_resume/stages/C_public/rows.jsonl	scalar_selects:COne

Key	Value	Source	Field or derivation
tabPPscalarselectsC1	9x6	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	scalar_selects:C1
tabPPscalarselectsDi	9x6	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	scalar_selects:Di
tabPPscalarselectsJ	50	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	scalar_selects:J
tabPPscalarselectsSer	20x16	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	scalar_selects:Ser
tabPPvectoraxpyC0ne	11x16	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	vector_axpy:C0ne
tabPPvectoraxpyC1	10x32	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	vector_axpy:C1
tabPPvectoraxpyDi	10x24	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	vector_axpy:Di
tabPPvectoraxpyJ	176	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	vector_axpy:J
tabPPvectoraxpySer	14x73	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	vector_axpy:Ser
tabPPvectorbitmixC0ne	13x24	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	vector_bitmix:C0ne
tabPPvectorbitmixC1	10x40	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	vector_bitmix:C1
tabPPvectorbitmixDi	10x40	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	vector_bitmix:Di
tabPPvectorbitmixJ	312	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	vector_bitmix:J
tabPPvectorbitmixSer	16x89	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	vector_bitmix:Ser
tabPPvectorreductionC One	12x32	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	vector_reduction:C0ne
tabPPvectorreductionC 1	12x40	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	vector_reduction:C1
tabPPvectorreductionD i	12x40	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	vector_reduction:Di
tabPPvectorreductionJ	384	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	vector_reduction:J
tabPPvectorreductionS er	19x137	p2/next_round_20260925/PUBLIC_SCORE. jsonandp2/third_round_20260925_res ume/stages/C_public/rows.jsonl	vector_reduction:Ser
tabPubSerial	1.000000	definition	scoreofserialagainstitse lf
weC1C	9	paper/generated/worked_example.js on(worked_example.compute)	methods.C1.C
weC1J	144	paper/generated/worked_example.js on(worked_example.compute)	methods.C1.J
weC1S	16	paper/generated/worked_example.js on(worked_example.compute)	methods.C1.S
weCapCcap	17	paper/generated/worked_example.js on(worked_example.compute)	caps.Ccap
weCapJ	240	paper/generated/worked_example.js on(worked_example.compute)	caps.J0
weCapLC	9	paper/generated/worked_example.js on(worked_example.compute)	caps.LC
weCapLS	8	paper/generated/worked_example.js on(worked_example.compute)	caps.LS

Key	Value	Source	Field or derivation
weCapScap	26	paper/generated/worked_example.json(worked_example.compute)	caps.Scap
weCatalogue	11	paper/generated/worked_example.json(worked_example.compute)	len(catalogue)
weChildCerts	3	paper/generated/worked_example.json(worked_example.compute)	len(child_certificates)
weChildOp	0	paper/generated/worked_example.json(worked_example.compute)	child_decision[0]
weChildVal	2	paper/generated/worked_example.json(worked_example.compute)	child_decision[1]
weClassicalC	9	paper/generated/worked_example.json(worked_example.compute)	methods.classical.C
weClassicalJ	153	paper/generated/worked_example.json(worked_example.compute)	methods.classical.J
weClassicalS	17	paper/generated/worked_example.json(worked_example.compute)	methods.classical.S
weCombDeclared	7,873,200	paper/generated/worked_example.json(worked_example.compute)	combinations_declared
weCombRoot	972,000	paper/generated/worked_example.json(worked_example.compute)	combinations_root
weDirectphase1C	10	paper/generated/worked_example.json(worked_example.compute)	methods.direct_phase1.C
weDirectphase1J	240	paper/generated/worked_example.json(worked_example.compute)	methods.direct_phase1.J
weDirectphase1S	24	paper/generated/worked_example.json(worked_example.compute)	methods.direct_phase1.S
weEdges	7	paper/generated/worked_example.json(worked_example.compute)	edges_total
weEpochs	4	paper/generated/worked_example.json(worked_example.compute)	epochs.count
weHorizon	17	paper/generated/worked_example.json(worked_example.compute)	horizon
weImprovements	3	paper/generated/worked_example.json(worked_example.compute)	len(epochs.steps)
weLooksC	3	paper/generated/worked_example.json(worked_example.compute)	child_edge_looks.C1
weLooksR	14	paper/generated/worked_example.json(worked_example.compute)	child_edge_looks.R0
weNodes	5,476	paper/generated/worked_example.json(worked_example.compute)	epochs.nodes
weOps	8	paper/generated/worked_example.json(worked_example.compute)	len(operations)
wePubC1Fullcycles	9	p2/third_round_20260925_resume/stages/C_public/rows.jsonl	public_row
wePubC1Fullscratch	16	p2/third_round_20260925_resume/stages/C_public/rows.jsonl	public_row
wePubC1Shortcycles	10	p2/third_round_20260925_resume/stages/C_public/rows.jsonl	public_row
wePubC1Shortscratch	16	p2/third_round_20260925_resume/stages/C_public/rows.jsonl	public_row
wePubR0Fullcycles	9	p2/third_round_20260925_resume/stages/C_public/rows.jsonl	public_row
wePubR0Fullscratch	16	p2/third_round_20260925_resume/stages/C_public/rows.jsonl	public_row
wePubR0Shortcycles	10	p2/third_round_20260925_resume/stages/C_public/rows.jsonl	public_row
wePubR0Shortscratch	17	p2/third_round_20260925_resume/stages/C_public/rows.jsonl	public_row
weQAChosen	16	paper/generated/worked_example.json(worked_example.compute)	addressqueryvgate:decodedaddress
weQAhi	16	paper/generated/worked_example.json(worked_example.compute)	addressqueryvgate:rangehi
weQTChosen	1	paper/generated/worked_example.json(worked_example.compute)	schedule_queries[2].chosen_index
weQThi	7	paper/generated/worked_example.json(worked_example.compute)	schedule_queries[2].intervals[0].hi
weQTlo	0	paper/generated/worked_example.json(worked_example.compute)	schedule_queries[2].intervals[0].lo
weQTop	2	paper/generated/worked_example.json(worked_example.compute)	indexofthedisplayschedulequery(operationid)
weQTsurv	3	paper/generated/worked_example.json(worked_example.compute)	len(schedule_queries[2].cubes)
weQone980026C	23	paper/generated/worked_example.json(worked_example.compute)	q1.980026.first_CSJ[0]

Key	Value	Source	Field or derivation
weQone980026J	1,840	paper/generated/worked_example.json(worked_example.compute)	q1.980026.first_CSJ[2]
weQone980026S	80	paper/generated/worked_example.json(worked_example.compute)	q1.980026.first_CSJ[1]
weQone980183C	22	paper/generated/worked_example.json(worked_example.compute)	q1.980183.first_CSJ[0]
weQone980183J	968	paper/generated/worked_example.json(worked_example.compute)	q1.980183.first_CSJ[2]
weQone980183S	44	paper/generated/worked_example.json(worked_example.compute)	q1.980183.first_CSJ[1]
weQueries	15	paper/generated/worked_example.json(worked_example.compute)	first_answer_queries.ques
weRootCerts	5	paper/generated/worked_example.json(worked_example.compute)	len(root_certificates)
weScaleFinalJ	384	paper/generated/worked_example.json(worked_example.compute)	scale.final_CSJ[2]
weScaleFirstJ	480	paper/generated/worked_example.json(worked_example.compute)	scale.first_CSJ[2]
weScaleNodes	9,999	paper/generated/worked_example.json(worked_example.compute)	scale.nodes
weScaleOps	19	paper/generated/worked_example.json(worked_example.compute)	scale.operations
weSerialC	11	paper/generated/worked_example.json(worked_example.compute)	methods.serial.C
weSerialJ	462	paper/generated/worked_example.json(worked_example.compute)	methods.serial.J
weSerialS	42	paper/generated/worked_example.json(worked_example.compute)	methods.serial.S
weStarterC	11	paper/generated/worked_example.json(worked_example.compute)	methods.starter.C
weStarterJ	462	paper/generated/worked_example.json(worked_example.compute)	methods.starter.J
weStarterS	42	paper/generated/worked_example.json(worked_example.compute)	methods.starter.S
weStarterScore	1.000000	paper/generated/worked_example.json(worked_example.compute)	starter_public.score(starter=serialonall8)
weStepAC	10	paper/generated/worked_example.json(worked_example.compute)	epochs.steps[0].to[0]
weStepAS	17	paper/generated/worked_example.json(worked_example.compute)	epochs.steps[0].to[1]
weStepBC	9	paper/generated/worked_example.json(worked_example.compute)	epochs.steps[1].to[0]
weStepBS	17	paper/generated/worked_example.json(worked_example.compute)	epochs.steps[1].to[1]
weStepCC	9	paper/generated/worked_example.json(worked_example.compute)	epochs.steps[2].to[0]
weStepCS	16	paper/generated/worked_example.json(worked_example.compute)	epochs.steps[2].to[1]
weSubC	4,440	paper/generated/worked_example.json(worked_example.compute)	subtree[-1].looks_C1
weSubCerts	1,655	paper/generated/worked_example.json(worked_example.compute)	subtree[-1].certificates
weSubChildren	3,001	paper/generated/worked_example.json(worked_example.compute)	subtree[-1].children
weSubR	27,874	paper/generated/worked_example.json(worked_example.compute)	subtree[-1].looks_R0
weSubRatio	0.16	paper/generated/worked_example.json(worked_example.compute)	subtree[-1].looks_C1/looks_R0
weTimeC1	100.4	p2/third_round_20260925_resume/stages/C_public/rows.jsonl	public_median_seconds
weTimeC1Full	233.5	p2/third_round_20260925_resume/stages/C_public/rows.jsonl	public_median_seconds
weTimeC1Short	10.4	p2/third_round_20260925_resume/stages/C_public/rows.jsonl	public_median_seconds
weTimeClassical	0.154	results/direct_index_v4_optimization_repair2/lead_review/final/runs.json	phase1_median_seconds
weTimeDirect	441.680	results/direct_index_v4_optimization_repair2/lead_review/final/runs.json	phase1_median_seconds
weTimeR0	100.5	p2/third_round_20260925_resume/stages/C_public/rows.jsonl	public_median_seconds
weTimeR0Full	239.7	p2/third_round_20260925_resume/stages/C_public/rows.jsonl	public_median_seconds

Key	Value	Source	Field or derivation
weTimeROShort	10.4	p2/third_round_20260925_resume/stages/C_public/rows.jsonl	public_median_seconds
weTimeSerial	0.015	results/direct_index_v4_optimization_repair2/lead_review/comparison/runs.json	phase1_median_seconds

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